FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 05, 19 – 23, 37 - 41)*** *contains 15 multiple choice questions which have* ***one correct*** *answer. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (06 – 13, 24 – 31, 42 - 49) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

2. ***Section-C (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

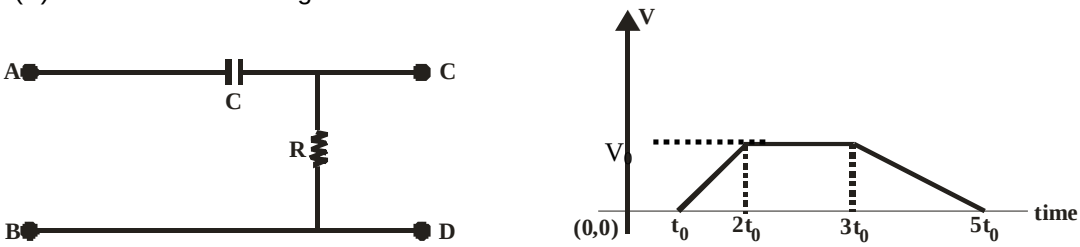
Physics

PART – I

SECTION – A
(One Options Correct Type)

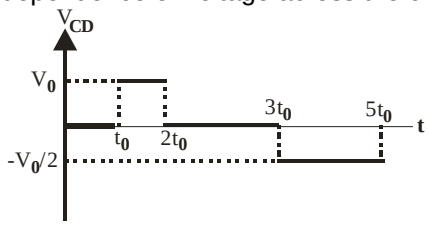
This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A varying voltage is applied to the clamps AB such that the voltage across the capacitor plates (V) varies as shown in figure

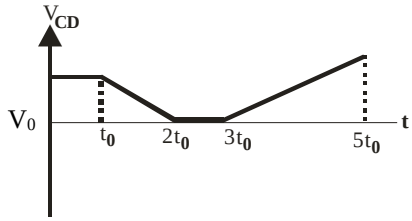


Time dependence of voltage across the clamp CD :

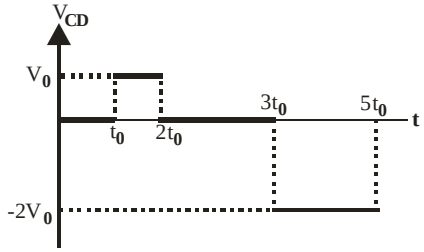
(A)



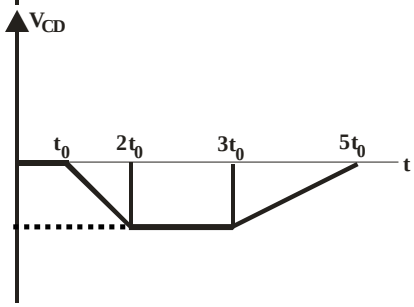
(B)



(C)



(D)



Space for Rough work

2. A block A of mass m and a box B of same mass are lying on a horizontal surface as shown in the figure. The coefficient of kinetic friction between the block and horizontal surface is μ_A, while that between the box and surface is μ_B. Given that $\mu_B > \mu_A$. At

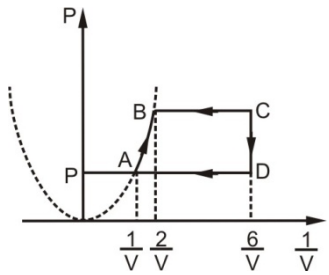


t = 0, the position is as shown in the figure and both are simultaneously given a velocity v_0 to the right. The time after which block will collide with the right end of the box is

(A) $\sqrt{\dfrac{2(L-\ell)}{(\mu_B-\mu_A)g}}$ (B) $\sqrt{\dfrac{2(L+\ell)}{(\mu_B+\mu_A)g}}$

(C) $\sqrt{\dfrac{4(L+\ell)}{(\mu_B-\mu_A)g}}$ (D) $\sqrt{\dfrac{4(L-\ell)}{(\mu_B-\mu_A)g}}$

3. One mole of an ideal monatomic gas is taken through a thermodynamic cycle ABCDA as shown in the figure. The process AB is a part of a parabola. Whose vertex is at the origin.



$\left(C_V = \dfrac{3}{2}R\right)$

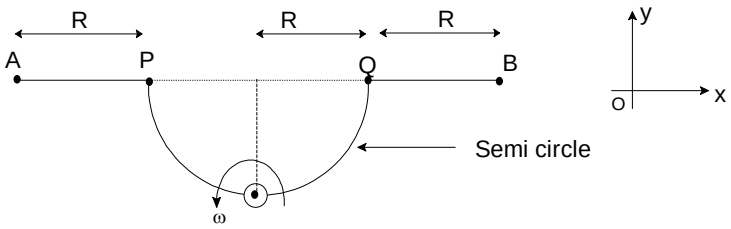
The total external work done on the gas in the whole process ABCDA is

(A) $-\dfrac{3}{2}PV$ (B) $\dfrac{3}{2}PV$

(C) $\dfrac{5}{6}PV$ (D) $-\dfrac{7}{6}PV$

Space for Rough work

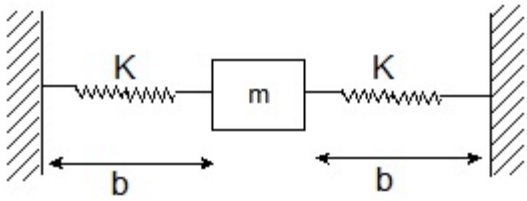
4. A conductor AB is rotated about O as shown in the figure such that its axis of rotation is parallel to the magnetic field $\vec{B} = B_0\hat{k}$ with uniform angular speed ω_0. The potential difference between points P and Q is



(A) $2B\omega_0R^2$ (B) $4B\omega_0R^2$

(C) zero (D) $B\omega_0\pi^2R^2$

5 In the adjacent figure, a block of mass m is rigidly attached to two identical spring of stiffness K each. The other ends of the spring are connected to the fixed wall. When the block is in equilibrium, length of each spring is b, which is greater than the natural length ℓ of the spring. The Time – period of the oscillation of the block if it is displaced by small distance X perpendicular to the length of the spring and released. (neglect gravity)



(A) $2\pi\sqrt{\dfrac{mb}{k(b-\ell)}}$ (B) $2\pi\sqrt{\dfrac{mb}{2k(b-\ell)}}$

(C) $2\pi\sqrt{\dfrac{m(b-\ell)}{kb}}$ (D) $2\pi\sqrt{\dfrac{m(b-\ell)}{2kb}}$

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

6. A dielectric slab of thickness '*d*' is inserted in the parallel plate capacitor whose negative plate is at $x = 0$ and positive plate is at $x = 3d$. The slab is equidistant from the plates. The capacitor is given some charge. As *x* goes from 0 to 3*d*.
(A) the magnitude of the electric field remains the same
(B) the direction of the electric field remains the same
(C) the electric potential increases continuously
(D) the electric potential increases first, then decreases and again increases

Space for Rough work

7. The wavelength of k_α x-rays for lead isotopes P_b^{208}, P_b^{206}, P_b^{204} and λ_1, λ_2 and λ_3 respectively. Then

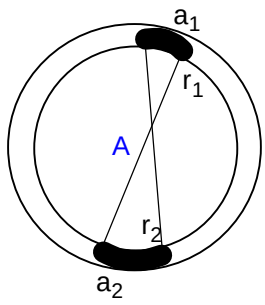
(A) $\lambda_1 = \lambda_2 = \lambda_3$ (B) $\lambda_1 > \lambda_2 > \lambda_3$

(C) $\lambda_1 < \lambda_2 < \lambda_3$ (D) $\lambda_2 = \sqrt{\lambda_1 \lambda_3}$

8. Radiations of monochromatic waves of wavelength 400 nm are made incident on the surface of metals Zn, Fe and Ni of work functions 3.4 eV, 4.8 eV and 5.9 eV respectively (take hc = 12400 eV- A^o) :

(A) Maximum KE associated with photoelectrons from the surface of any metal is 0.3 eV.
(B) No photoelectrons are emitted from the surface of Ni.
(C) If the wavelength of source of radiation is doubled then KE of photoelectrons is also doubled.
(D) Photoelectrons will be emitted from the surface of all the three metals if the wavelength of incident radiations is less than 200 nm

9. A wire having a uniform linear charge density λ, is bent in the form of a ring of radius *R.* Point *A* as shown in the figure, is in the plane of the ring but not at the centre. Two elements of the ring of lengths a_1 and a_2 subtend very small same angle at the point A. They are at distances r_1 and r_2 from the point A respectively. Then for $r_1 < r_2$, choose the correct option(s).



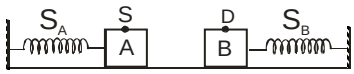
(A) the ratio of charge of elements a_1 and a_2 is $\frac{r_1}{r_2}$

(B) the elements a_1 produces greater magnitude of electric field at A than elements a_2.

(C) the elements a_1 and a_2 produce same potential at A.

(D) the direction of net electric field at A is towards element a_1.

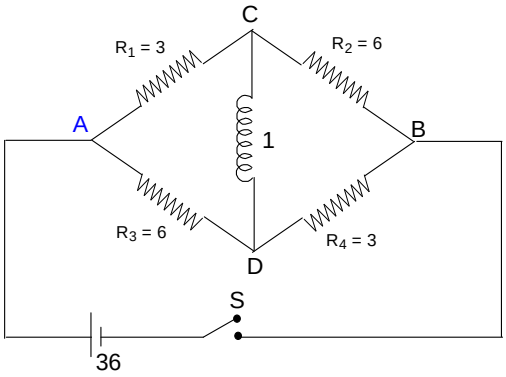
10. A source emitting a sound of frequency 300 Hz is fixed on block 'A' while detector is fixed on block B detects the sound. The blocks 'A' and 'B' are simultaneously displaced towards each other through a distance 1 m and then left to vibrate. Velocity of sound in air = 340 m/sec and S_A and S_B are identical. The frequency of vibration of S_A is 2 Hz.



(A) Maximum velocity of detector is 4πm / sec
(B) Maximum frequency detect by detector is 323 Hz.
(C) Minimum frequency detect by detector is 278 Hz
(D) Phase difference between S and D is $\pi / 2$

Space for Rough work

11. The switch S is closed at t = 0 as shown in the figure.



(A) the current as function of time in the inductor is $3(1 - e^{-4t})$A.
(B) the current through R_1 just after closing of switch is 8A
(C) the power supply by the battery as function of time is $36(9 - e^{-4t})$
(D) the current through R_2 after long time closing of switch is 3A

12. In a certain series LCR A.C. circuit. It is found that $X_L = 2X_C$ and phase difference between the current and voltage is $\frac{\pi}{4}$. If now capacitance is made $1/4^{th}$ & the inductance and resistance are doubled then which are not correct.

(A) Voltage will lead current by a phase of $\tan^{-1}\left(\frac{1}{2}\right)$

(B) Voltage will lay behind current by a phase of $\tan^{-1}\left(\frac{1}{2}\right)$

(C) Voltage will lag behind current by a phase of $\frac{\pi}{4}$

(D) Voltage and current will be in same phase

13. Two concentric metallic shell's of radius R and 2R, out of which the inner shell is having Q and outer shell is uncharged. If they are connected with a conducting wire. Then,
(A) Q amount of charge will flow from inner to outer shell
(B) Q/e no. of electrons will flow from outer to inner shell
(C) $\frac{KQ^2}{4R}$ amount of heat is produced in the wire
(D) $\frac{KQ^2}{2R}$ amount of heat is produced in the wire

Space for Rough work

SECTION – C
(Single digit integer type)

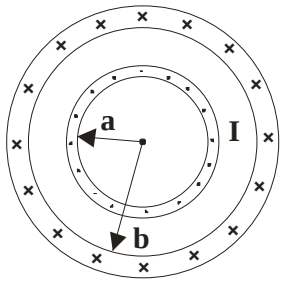
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. Find the energy E required (in MeV) to remove an alpha particle from $^{14}_{7}N$. The excess of mass of atom (in amu) is given below. (take 1 amu = 931 MeV)

Atom	(Mass of atom – A. amu)
$^{4}_{2}He$	0.00260
$^{10}_{5}B$	0.01294
$^{14}_{4}N$	0.00307

Fill E/2 in OMR sheet after rounding off to nearest integer.

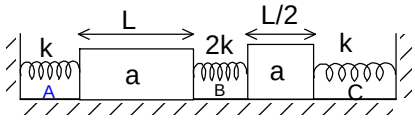
15. A thin hollow cylinder of radius a is surrounded co-axially by another hollow cylinder of radius b where b > a. Both the cylinder form part of an electric circuit and a current I flows through them as shown in figure.



Magnitude of pressure exerted on inner cylinder is $\frac{\mu_0 I^2}{K\pi^2 a^2}$.

Find K?

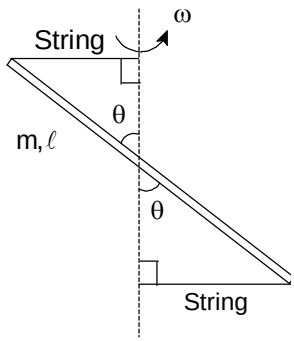
16. Two rigid rods of same material and of length L and $\frac{L}{2}$ are placed in equilibrium on a smooth horizontal plane at temperature T as shown in figure. Springs are in their natural length. If the temperature is increased by ΔT then the amount of energy stored in springs is $\frac{9}{5\beta}k\alpha^2L^2\Delta T^2$.



Where *k* is the spring constant of spring (*A*) and α is the coefficient of linear expansion of material. Then find the value of β (in proper units) (Note: Neglect the change in spring constant due to increase in temperature).

Space for Rough work

17. A stick of mass m and length ℓ spins with frequency ω around an axis, as shown in figure. The stick makes an angle θ with the axis and is pivoted at its centre. It is kept in this motion by two strings which are perpendicular to the axis. Then tension in the string is $T = \frac{K}{24} m\ell\omega^2 \sin\theta$. Find K?



18. A spherical shell of radius R is filled with water. Temperature of atmosphere is $(-\theta)^0$ C, where θ is a positive constant. The shell is exposed to atmosphere and all water comes down to 0^0C and then its starts freezing from outer surface towards the centre of the shell. Assuming shell to be surface towards the centre of the shell. Assuming shell to be highly conducting, if time for whole mass of water at 0^0C to freeze is $t = \frac{\delta L R^2}{\alpha K T}$. Thermal conductivity of ice is K and latent of fusion is L. Density of water is δ. Neglect expansion during fusion. Find α?

Space for Rough work

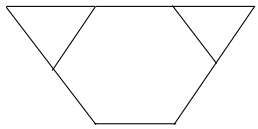
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Correct statements regarding 'X' is:



(A) 2 chiral centre and 2 stereo isomer
(B) 4 chiral centre and 3 stereo isomer
(C) 4 chiral centre and 2 stereo isomer
(D) 2 chiral centre and 3 stereo isomer

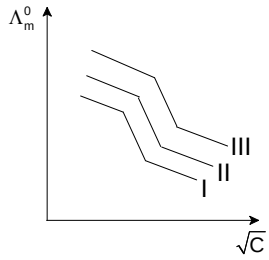
20. A photon is emitted in $\beta-$ line of Balmer series in He^+. This photon is absorbed by an electron present in ground state of H – atom. If uncertainty in position is equal to De-Broglie wavelength then uncertainty in De-Broglie wavelength is:

(A) $0.529\,\overset{0}{A}$ (B) $2.116\,\overset{0}{A}$
(C) $4.232\,\overset{0}{A}$ (D) $0.265\,\overset{0}{A}$

21. For which of the following complex / ion total number of geometrical isomers is equal to total number of optical isomers?

(A) $\left[IrBr(en)(H_2O)_2\right]^+$ (B) $\left[CoBr_2(en)_2\right]^+$
(C) $\left[PtBr(NO_2)(SCN)(NH_3)_3\right]$ (D) None of these

22. Below three graphs are for $R(CH_2)_nCOONa$, correct value of 'n' may be correctly matched as:



	I	II	III
(A)	16	12	8
(B)	8	12	16
(C)	12	8	16
(D)	12	12	12

Space for Rough work

23. $FeS + H_2SO_4 \longrightarrow B$

which is **incorrect** about gas 'B' ?

(A) It produces white ppt. with $FeCl_3$

(B) It produces black ppt. with $(CH_3COO)_2Pb$.

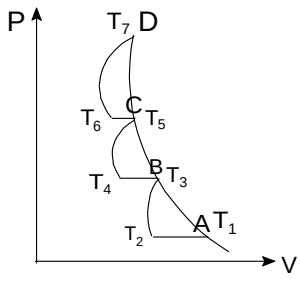
(C) It decolourises the purple colour of $KMnO_4$.

(D) It produces purple colour solution with sodium nitroprusside

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

24. Through a number of isobaric cooling and adiabatic compression 1 mol of helium (ideal gas) is compressed from 1 bar, 300 K to 32768 bar, 300 K. Let the process be accomplished in three steps (AB, BC and CD). At each step helium is first cooled and then compressed. At the end of each step the pressure increases 32 times and temperature returns to original value of 300 K. Which of the following is/are correct? $[R = 2 cal/mol/K]$



(A) $T_1 = T_3 = T_5 = T_7$

(B) $T_2 = T_4 = T_6$

(C) $Q_{total} = -3375 cal$

(D) $\Delta S_{system} = 15R \ln 2$

25. 34.3 g Pb_3O_4 is dissolved in 500 ml of 4 M HNO_3 then (Pb = 207)

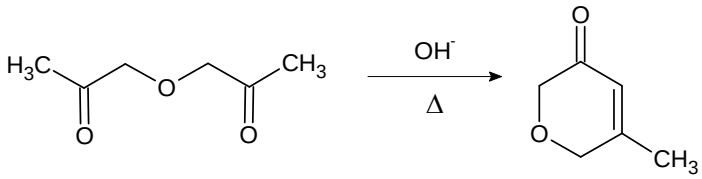
(A) there is no residue left

(B) the wt. of residue is 11.95 g

(C) 300 ml of 6 M NaOH is required to neutralise excess of HNO_3

(D) It is a redox reaction

26.



Which of the following step, mechanism is/are involved in above conversion?

(A) Acid base

(B) Nucleophilic addition

(C) E_1CB

(D) dehydration

Space for Rough work

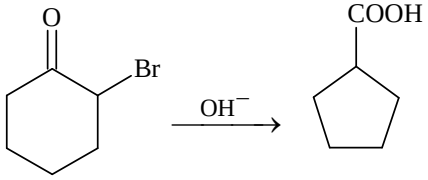
27. Select correct statement(s) for H_2O_2.
(A) It is prepared by air oxidation of 2-ethylanthraquionone
(B) It can reduce I_2 into I^- (aq) in alkaline medium
(C) It can act as reductant with $KMnO_4$
(D) Its melting point is lower than that of H_2O

28. Which of the following are disproportionation reactions?

(A) $2PhCHO \xrightarrow{Al(OEt)_3} Ph-\overset{\overset{O}{\|}}{C}-OCH_2Ph$

(B) $2HCHO \xrightarrow{OH^-} CH_3OH + HCOO^-$

(C) $2CH_3CHO \xrightarrow{OH^-} CH_3-\underset{\underset{OH}{|}}{CH}-CH_2CHO$

(D) 

29. Select the incorrect choice from the following.
(A) ΔS_{mix} for all types of solutions (ideal & non-ideal solutions) is positive.
(B) ΔG_{mix} of ideal solutions is always negative
(C) Partially miscible solutions with $\Delta H_{mix} = +ve$ value, are completely miscible at upper consulate temperature at which $\Delta G_{mix} = -ve$
(D) Azeotrope contains only one component formed as a result of definite union of the two liquids and the composition of azeotrope does not change with pressure. (consider binary liquids)

Space for Rough work

30. Which among the following will give cinnamic acid?

(A) $Ph-CHO \xrightarrow[(2)H_3O^+]{(1)Ac_2O\ /\ AcONa,\ heat}$

(B) $Ph-CHO \xrightarrow[(2)H_3O^+\ (3)\Delta]{(1)CH_2\left(COOC_2H_5\right)_2,\ Pyridine}$

(C) $Ph-CHO \xrightarrow[(2)H_3O^+\ (3)\Delta]{BrCH_2COOEt + Zn + \Delta}$

(D) $Ph-CHO \xrightarrow[(2)H_3O^+]{(1)HCN}$

31. In normal spinel having one eighth of tetrahedral holes occupied by one type of metal ion and one half of the octahedral holes occupied by another type of metal ion and oxide ion forms F.C.C. lattice. Such spinel is formed by Mg^{2+}, Al^{3+}, O^{2-}. If oxide ion is replaced by $X^{-8/3}$ ion then:

(A) The number of anionic vacancies per unit cell is 2
(B) Formula of spinel becomes $MgAl_2X_3$
(C) Spinel becomes charged species
(D) Al^{3+} is in octahedral void

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. Suppose that at time 't' the state function of a one – particle system is:

$\psi = \left[2/\left(\pi C^2\right)\right]^{3/4} e^{-\left(x^2+y^2+z^2\right)/C^2}$ where C = 2 nm

The probability of finding the particle at any time in the tiny cubic region with its centre at x = 1.2 nm, y = -1 nm, z = 0 nm and with edges of each of length 0.004 nm is $X \times 10^{-Y}$ (in scientific notation), find Y. $\left[e^{-1.22} = 0.30,\ (1/2\pi)^{3/2} = 0.06\right]$

Space for Rough work

33. 12.5 milli moles of NH_4NO_3 is dissolved in enough water to make 25 ml of solution. The initial temperature is 25.8 ^{0}C and final temperature after solid dissolves is 21.8^{0}C. Calculate the enthalpy of solution for NH_4NO_3(s) in kcal/mol.
Give: $d_{solution}$ = 1 g/ml, specific heat capacity of solution = 1 cal/g/K

34. A hydrocarbon (X) contains 91.2% carbon and 8.8% hydrogen. The compound on chlorination using $Cl_2 / h\nu$ and $Cl_2 / AlCl_3$ gives three isomeric monochloro substituted products, then the value of $\frac{\text{Total number of atoms present in (X)}}{5} = ?$

35. Consider the following compounds: Na_2SO_3, NaCl, $Na_2C_2O_4$, Na_2HPO_4, $NaNO_2$, Na_2CrO_4, CH3COONa,
when treated separately with $AgNO_3$ solution, how many of them give a white precipitate?

36. If certain decomposition reaction found to obey equation in term of partial pressure (P) of reactant as:
$\frac{a - P^2}{P^2} = bkt$, what is order of reaction if a, b and k are constant (t represents time) ?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. ABC and BDE are two equilateral triangles such that D is the mid-point of BC, AE intersect BC at F. If $2|z-z_d|=|z_a-z_c|$ and $\left(\frac{|z-z_a|_{max}}{|z-z_a|_{min}}-2\right)^2=k\frac{\text{area}(\Delta FED)}{\text{area}(\Delta AFC)}$ then value of k is

{where z represents complex number, z_a, z_d & z_c are the affixes of A, D & C respectively}

(A) 21 (B) 22
(C) 23 (D) 24

38. The sum of series $\sum_{r=1}^{100}\left(\frac{1}{\cos\theta+\cos(2r+1)\theta}\right)$ is

(A) $\left(\frac{\sin(101\theta)}{\sin\theta.\cos(100\theta)}\right)$ (B) $\left(\frac{\sin(100\theta)}{\sin2\theta.\cos(101\theta)}\right)$

(C) $\left(\frac{\tan(101\theta)}{\sin(100\theta)}\right)$ (D) none of these

39. If line through P (a, 2) meets the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at A and D and meets the axes at B and C, so that PA, PB, PC, PD are in G. P., then

(A) $a\in[5,\infty)$ (B) $a\in R$
(C) $a\in(-\infty,-6)\cup[6,\infty)$ (D) none of these

Space for Rough work

40. If , $\sum_{k=1}^{n}(k^2+5k+7)(k+2)! + 3.3! = 2003.2003!$ then the value of n is,

(A) 1000 (B) 2001
(C) 2002 (D) None of these

41. The value of $\int_0^{1/2} e^x\left(\sin^{-1}x - \frac{x}{(1-x^2)^{3/2}}\right)dx$ is

(A) $\frac{\sqrt{e}\pi}{6} - \frac{2}{3\sqrt{3}}$ (B) $\sqrt{e}\left(\frac{\pi}{6} - \frac{2}{\sqrt{3}}\right)+1$

(C) $\frac{\sqrt{e}\pi}{6} - \frac{4}{3\sqrt{3}}$ (D) 0

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

42. If f(x) be a continuous function for all real values of x and satisfies $\int_3^x f(t)dt = \frac{x^2}{2} + \int_x^8 t^2 f(t)dt,$

then

(A) $\int_{-1/2}^{1} f(x)dx$ is equal to $\ln\left(\frac{5}{8}\right)$ (B) $\int_{-1/2}^{1} f(x)dx$ is equal to $\frac{1}{2}\ln\left(\frac{8}{5}\right)$

(C) $f(x) = x - x^2 f(x)$ (D) $f(x) = x + x^2 f(x)$

43. A circle of radius 4 cm is inscribed in $\Delta ABC,$ which touches the side BC at D. if $BD = 6cm$: $DC = 8cm$ then

(A) The triangle is necessarily acute angled Δ

(B) $\tan\frac{A}{2} = \frac{4}{7};$

(C) Perimeter of the triangle ABC is 42 cm

(D) Area of ΔABC is 84 cm^2

Space for Rough work

44. If a, b, c $\in$ Q then which of the following equations has rational roots

(A) $ax^2 + bx + c = 0$, Where if a + b + c = 0

(B) $(a + c - b)x^2 + 2cx + (b + c - a) = 0$

(C) $abc^2x^2 + 3a^2cx + b^2cx - 6a^2 - 4ab - 2b^2 = 0$

(D) $(a + b - c)x^2 + (a + c - b)x + (b + c - a) = 0$

45. The value of $\int \frac{dx}{1 + e\cos x}$ must be same as

(A) $\frac{1}{\sqrt{1-e^2}} \tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\ \tan\frac{x}{2}\right) + c$ (e lies between 0 and 1)

(B) $\frac{2}{\sqrt{1-e^2}}\ \tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{x}{2}\right) + c$, (e lies between 0 and 1)

(C) $\frac{1}{\sqrt{e^2-1}} \log \frac{e + \cos x + \sqrt{e^2-1}\sin x}{1 + e\cos x}$ + c, (e is greater than 1)

(D) $\frac{2}{\sqrt{e^2-1}} \log \frac{e + \cos x + \sqrt{e^2-1}\sin x}{1 + e\cos x} + c$, (e is greater than 1)

46. A line passes through the points whose position vectors are $\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + \hat{k}$. The position vector of a point on it at a unit distance from the first point is/are

(A) $\frac{1}{5}\left(5\hat{i} + \hat{j} - 7\hat{k}\right)$

(B) $\frac{1}{5}\left(5\hat{i} + 9\hat{j} - 13\hat{k}\right)$

(C) $\frac{1}{5}\left(6\hat{i} + \hat{j} - 7\hat{k}\right)$

(D) $\frac{1}{5}\left(4\hat{i} + 9\hat{j} - 13\hat{k}\right)$

Space for Rough work

47. A tangent is drawn to the curve y = f(x) at P(x, y), cuts the x-axis at A. Now a perpendicular drawn from P(x, y) to x-axis meets the x-axis at B. If B is mid-point of OA (O being origin) and f (1) = 1, then

(A) General point on the curve is $\left(t, \frac{1}{t}\right)$, where t is a real parameter ($\neq 0$)

(B) y = f(x) is a circle

(C) Tangent at P(1, 1) is x + y = 2

(D) f ′(2) = –1/4

48. If $\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & bc & c^2+1 \end{vmatrix}$ = 1, where a, b, c are real, then which of the following is not true,

(A) $a^2+b^2+c^2=1$ (B) a + b + c = 1 (C) $a^2 + b^2 + c^2 = 0$ (D) a + b + c = 2

49. P is a point inside a ΔABC of area K (K>0) .The lengths of perpendiculars drawn to the sides BC, CA, AB of lengths a,b,c are respectively P_1, P_2, P_3.Then $\frac{a}{P_1}+\frac{b}{P_2}+\frac{c}{P_3}$ is minimum when

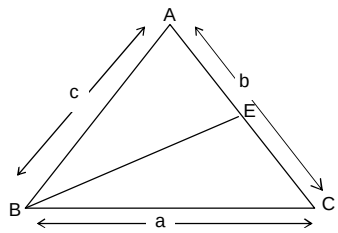
(A) P is incentre of ΔABC (B) P is equidistant to all the 3 sides

(C) $P_1=P_2=P_3=\frac{2K}{a+b+c}$ (D) P is orthocentre of ΔABC

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. In figure, a, b, c are sides of a triangle ABC such that a > b > c. If lengths of its medians are in geometric progression with median BE = $\frac{1}{2}$, then the least integral value of $\sqrt{a^2+b^2+c^2}$ is?



Space for Rough work

51. Let p, q, r and s are the probabilities of raining at four different places at some fixed moment, if M and N are the maximum and minimum value of $3(p^2 + q^2 + r^2 + s^2) - 2(p + q + r + s) + 4$, then value of [M – N] is (where [.] represent GIF)

52. If the number of ways in which 21 identical objects can be distributed among 5 persons such that each can get at least one object and no two person can get equal number of objects is $(k+1)5!$, then the value of 'k' is?

53. If α, β are two distinct real roots of the equation $ax^3 + x - 1 - a = 0$, $(a \neq -1, 0)$, none of which is equal to unity, and the value of $\lim\limits_{x \to \frac{1}{\alpha}} \dfrac{(1 + a)x^3 - x^2 - a}{(e^{1 - \alpha x} - 1)(x - 1)}$ is $\dfrac{a\lambda(k\alpha - \beta)}{\alpha}$, then the value of $k\lambda$ is?

54. Let equation $x^3 + px^2 + qx - q = 0$, where $p, q \in R - \{0\}$ has 3 real roots α, β, γ in H.P. and the minimum value of $\dfrac{1}{\alpha^2} + \dfrac{1}{\beta^2} + \dfrac{1}{\gamma^2}$ is $\dfrac{k}{12}$ then the value of k is ?

Space for Rough work